PROCEDURE FOR ROCKING TEST (WERE MEARSUREMENT)

IN SLEWING BEARING FOR MARINE CRANE

1. General

The slewing bearing is an essential part of the crane and must be well maintained.

Over the years the slewing bearing will have some wear, and to be able to know if there is a need for changing the slewing bearing it is needed to keep a record of the wear.

And crane manufacturers generally recommend that “rocking tests” are carried out every six(6) month in accordance with their instructions, and that the results are recorded and monitored in order to ensure the wear tolerances remain within the limits permitted by the designers.

2. Measuring procedure

2.1 Measurement period.

The initial measurement should be taken when crane is installed at shipyard within two(2) months from crane installation.

The measurements should be taken regularly, every six(6) months.

2.2 Ship condition at measuring of slewing bearing.

When measurements are to be taken the ship shall have as little list/trim as possible.

Neither a load nor a cargo handling equipment should be attached to the hook.

2.3 Measuring

2.3.1 Reference points

Two(2) reference points to be machined/marked on the front and rear side of crane bottom plate, see fig.1, and to be used in future measuring.

The first time a measurement is to be taken four(4) reference points must be marked on the top flange of pedestal or lower surface of slewing bearing

(outer surface of inner ring)

These reference points shall always be used at any future measuring.

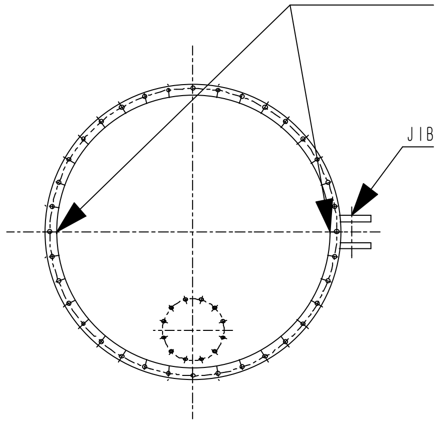


Fig. 1 Reference points of crane bottom plate

2.3.2 Procedure

Two measurements shall be taken on four positions of the slewing bearing, if possible, the jib pointing:

-Forward to the ship

-Starboard

-Aft

-Port

The measurement to be carried out by the use of sliding caliper or micrometer screw at the reference points marked.

It is important to use the same position for all further measurements to be able to compare the measurements.

For maximum permissible clearance(wear) ; See table 1 on page 3/5

※ WARNING

Do not enter the crane pedestal during crane operation.

Moving crane house and machine parts may suddenly cause body injury or death.

With the jib at maximum outreach two measurements are to be taken, see Fig.2.

The enclosed table is to be filled out as each Measurement is taken. It is important that a complete set of these measurements are made every 6 month. The measuring instruments to be used are to have an accuracy of at least 0.1mm. The first set of measurements are to be recorded upon completion of the installation of the crane or within two(2) months from crane installation.

If the wear measurement shows wear, grease samples can be taken for analyze.

See procedure on page 5/5.

The maximum allowable deviation between the initial set of measurements and all future measurements are indicated in the table. See table on page 3/5.

Should the wear rate suddenly show marked deviation from the expected rate, the measurement rate should be increased.

The safe functioning of the slew bearing is dangerously reduced if the maximum allowable deviation exceeded.

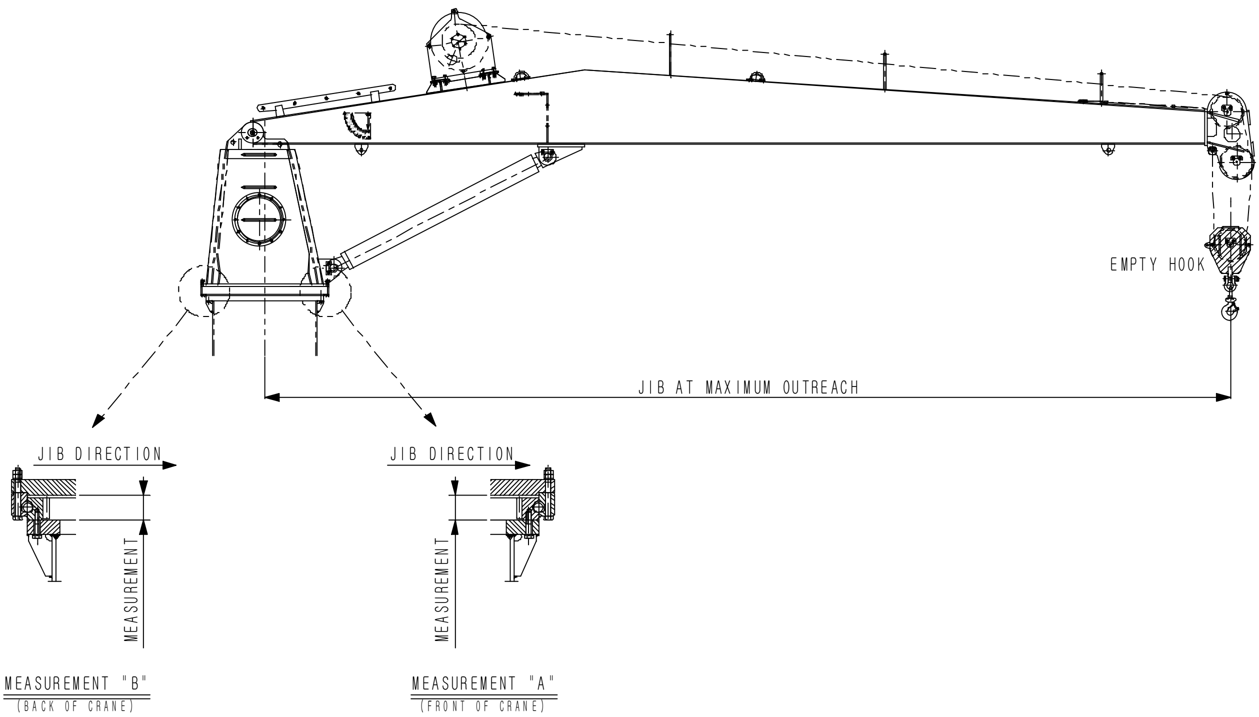


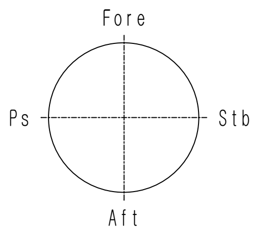
Fig. 2

Table 1. Maximum permissible clearance

Bearing Ball PCD (mm)	Ball diameter(mm)					
	25.4	28.575 31.75	34.925	38.1 39.688	44.45	47.625 50.8
	Max. allowable deviation in bearing clearance(mm)					
1250 and below	1.2	1.2	1.3	1.4	1.5	1.6
1500 and below	1.3	1.4	1.4	1.5	1.6	1.7
1750 and below		1.4	1.7	2.0	2.1	2.2
2000 and below		1.7	1.7	2.0	2.2	2.5
2250 and below			1.8	2.0	2.4	2.5
2500 and below			1.9	2.1	2.4	2.7

*Measurement and actual measurement. See table on page 4/5

MESURING OF WEAR ON SLEWING BEARING



VESSEL NAME :

CRANE No./Mfg.No :

Initial measurement (accuracy 0.1 mm)				Allowed wear (see table on page 4/6)
Date	Jib crane	A1(front of crane)	B1(Back of crane)	
	Fore			
	Stbd			
	Aft			
	Port			

Actual measurement(accuracy 0.1 mm)				Measure wear	
Date	Jib direction	A(front of crane)	B(Back of crane)	A-A1	B-B1
	Fore				
	Stbd				
	Aft				
	Port				

Actual measurement(accuracy 0.1 mm)				Measure wear	
Date	Jib direction	A(front of crane)	B(Back of crane)	A-A1	B-B1
	Fore				
	Stbd				
	Aft				
	Port				

PROCEDURE FOR GREASING SAMPLING

Following information is necessary in order to give a correct statement and allow further advice:

1. Type of grease used at lubrication(manufacturer and type)

2. Lubrication intervals(crane running or months)

3. Crane running hours.

4. Information where the samples are taken.

As a correct procedure of grease sampling is very important below procedure of should be followed.

PROCEDURE ;

- The grease sample can be taken either at the inner or outer seal of the bearing.

- One sample should be taken in the area under jib and one sample 180° in opposite direction.

- Clean the outer or inner seal close to the next grease nipple. When cleaning the area of the seal, prevent the cleaner from contacting the seals or form entering the raceway's system.

- Push new grease into the grease nipples/bearing without rotation and the first used grease which will come out at the seal.

Attention : Do not take fresh grease for analyzing !

The required quantity of grease for analyzing is approx. 2cm^3.

- The normal interval for grease sampling is 12 months.